

HYPERBUILDING PROGRAM

A VERTICAL ECOSYSTEM IN SANTIAGO



TYPE: BIM & Interoperability

YEAR: 2026

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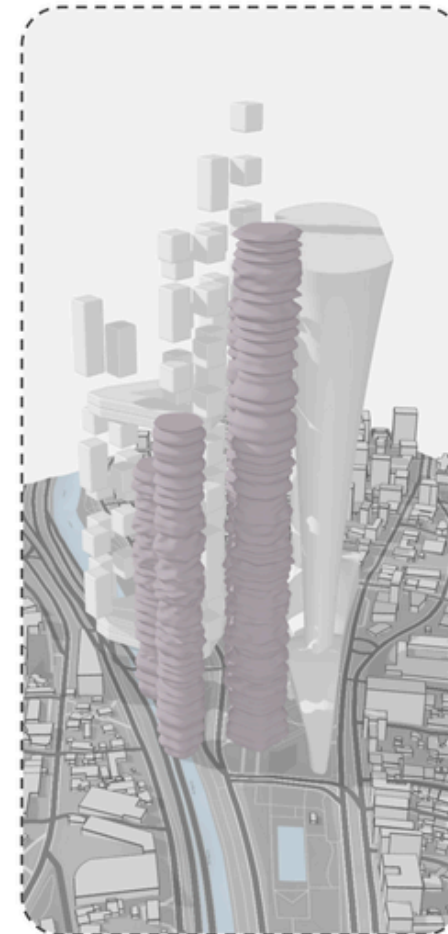
CONTEXT: IAAC Master in Advanced Computational Design (MACAD)

Hyperbuilding - The Cells is part of “*The Vertical Machine*”, an ambitious architectural complex comprising three massive towers of 1 million sqm each in Santiago, Chile. The project is developed by three specialized teams: Data, Structure and Program. As part of the Program Team, our goal was to coordinate with the other teams to define the massing strategy and to translate human needs into spatial logic, organizing the vertical distribution, and ensuring a livable and interconnected environment. We developed an automated pipeline to generate a program distribution based on both view quality and program interconnections.

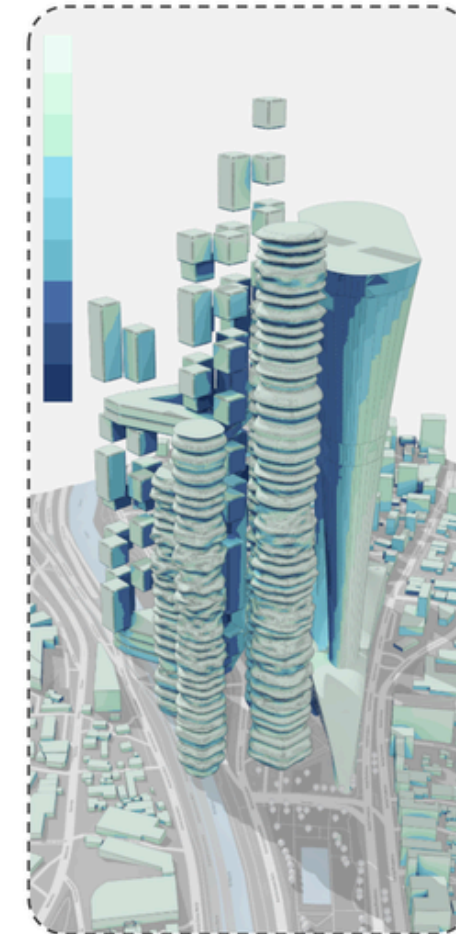
MASSING STRATEGY

Using key metrics from the early KPIs, several iterations were generated in Autodesk Forma and Ladybug, based on:

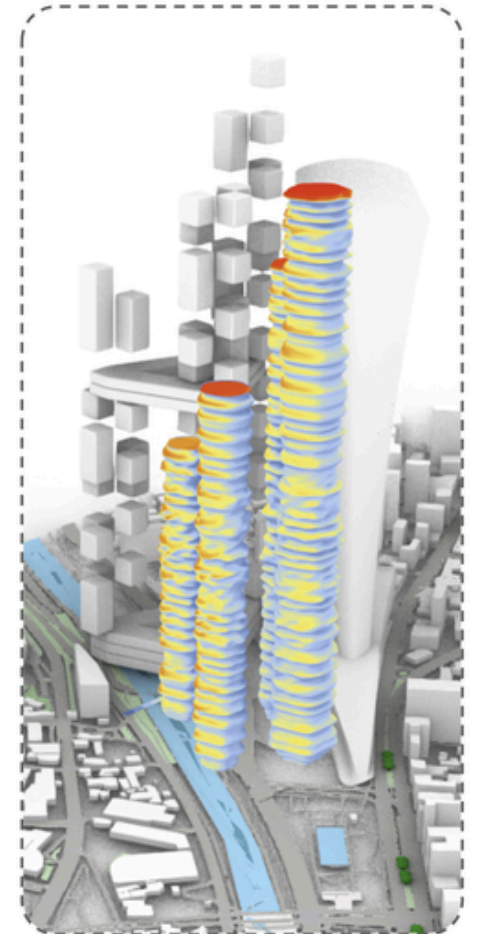
Shadow Study



Daylight Potential

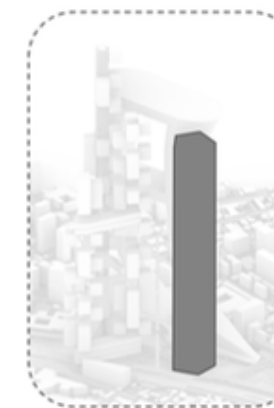


Sun Analysis



Initial Massing

Single Tower



Same Height Tower



Different Heights Tower



Data-Driven Massing

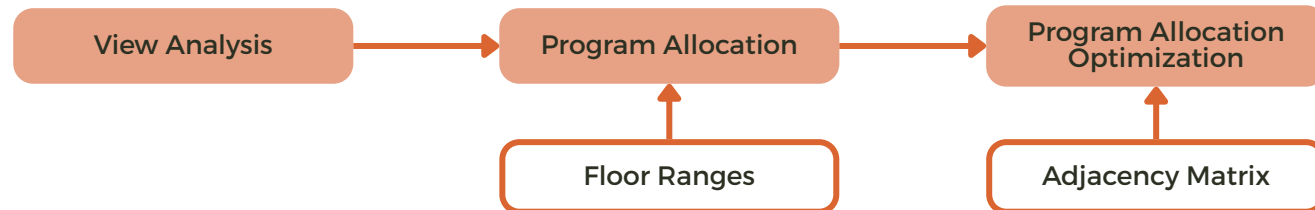
Bridges for green space



Final Massing

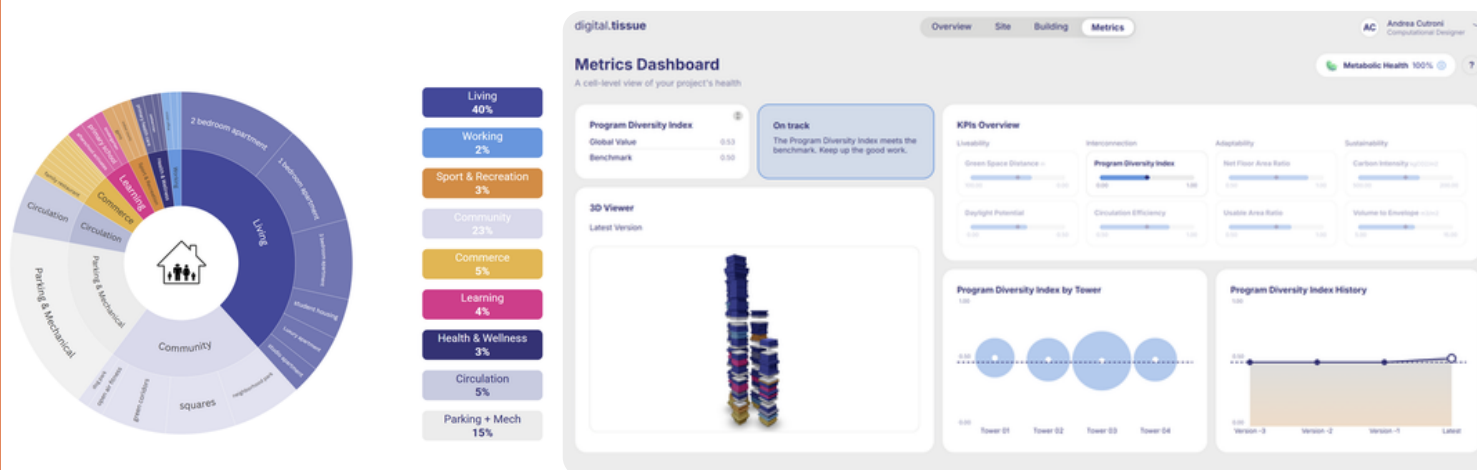


WORKFLOW



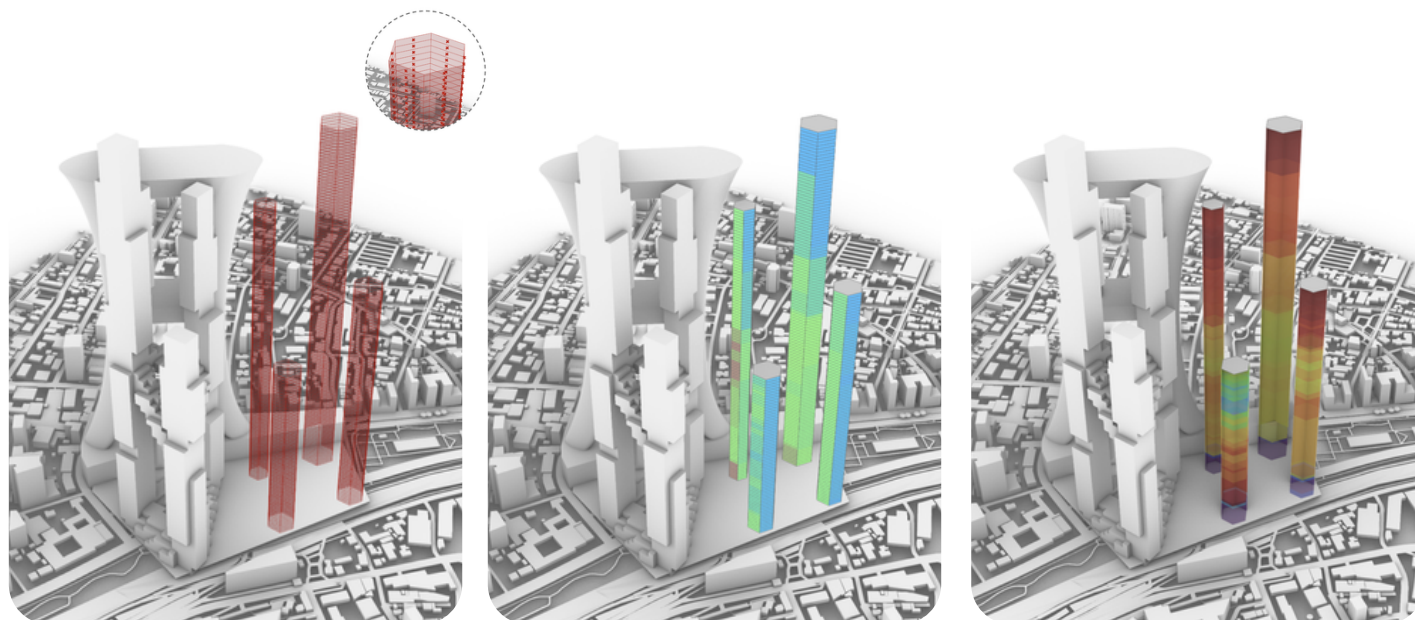
PROGRAM & INTEROPERABILITY

Data are sent to the metric dashboard to keep track of the KPIs and shared through Speckle, while a workflow via Rhino.inside.Revit enables real-time metrics and automated documentation.



VIEW ANALYSIS

View quality analysis informs the assignment of building programs to floors, establishing program ranges that guide the initial allocation strategy.



DATA-DRIVEN PROGRAM ALLOCATION

	LIVING	COMMUNITY	COMMERCE	LEARNING	WORKING	HEALTH	SPORT
LIVING	0	2	0	2	1	1	2
COMMUNITY	2	0	0	2	1	0	2
COMMERCE	0	0	0	1	2	1	1
LEARNING	2	2	1	0	1	0	2
WORKING	1	1	2	1	0	1	1
HEALTH	1	0	1	0	1	0	2
SPORT	2	2	1	2	1	2	0

0 = weak adjacency; 1 = neutral; 2 = strong adjacency

A Python script automates the program allocation optimization, applying a local optimum approach taking into account the adjacency matrix.

- Sport & Recreation
- Working
- Community
- Commerce
- Living
- Learning
- Health & wellness

